Shan Meshram

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.45
Jawahar Navodaya Vidyalaya, Barwani, Madhya Pradesh	Class XII (CBSE)	2024	91.20%
Jawahar Navodaya Vidyalaya, Balaghat, Madhya Pradesh	Class X (CBSE)	2022	96.20%

SCHOLASTIC ACHIEVEMENTS

Ranked in the **top 8%** nationwide among **250,000+** participants in the **AlgoUniversity Tech Fellowship (ATF) 2025-Stage 1**.

Vahani Scholarship - Selected among 50 scholars from ~3,000 applicants based on personal essays and two interview rounds.

PROJECTS

Handwritten Notes Knowledge Base -- RAG-based Notes Assistant | Python, Groq API, ChromaDB, Sentence-Transformers, Streamlit | GitHub

- Engineered an **AI-enabled RAG ingestion pipeline** using Vision LLMs (Llama 4 Scout via Groq API) to convert unstructured handwritten notes into structured Markdown, enforcing **prompt-engineered schema rules**.
- Developed an **NLP retrieval system** with local sentence-transformers (bge-small-en-v1.5) and ChromaDB vector database, implementing **top-k semantic search** to ground text LLMs (Llama 3.3 70B) with mandatory source citations to ensure **hallucination-free outputs**.
- Decoupled pipeline components into a **custom multi-model architecture**, optimizing API integration, latency, and compute overhead between vision extraction and query generation.
- Developed an **interactive web application** in Streamlit featuring dynamic retrieval controls and automated conversation summarization to transform user interactions into **reusable technology assets**.

Extended xv6 OS Kernel with Custom Scheduler, Virtual Memory & RAID Storage | C, RISC-V, xv6 | GitHub

- Engineered a custom system-level software solution in C, building a **Multi-Level Feedback Queue (MLFQ)** process scheduler with 4 priority levels and 128-tick global priority boosting to **eliminate CPU starvation**.
- Implemented **virtual memory management** featuring Clock page replacement with disk-backed swapping, and integrated simulated **fault-tolerant RAID 0, 1, and 5** storage sub-systems with XOR parity recovery.
- Developed **FCFS and SSTF disk I/O schedulers** supporting dynamic runtime policy switching via system calls to optimize seek latency while logging granular per-process metrics.

Multithreaded Sudoku Validator with Chunk & Round-Robin Scheduling | C, Pthreads | GitHub

- Designed a **parallel application in C** using POSIX threads (Pthreads), implementing dynamic **Chunk and Round-Robin** task-scheduling algorithms across matrix workload partitions.
- **Optimized application performance** by introducing early-termination logic to short-circuit thread execution upon error detection, reducing redundant CPU cycles and runtime latency.
- Built a high-resolution benchmarking and log-buffer **testing framework** to analyze thread contention, workload balancing, and **parallel speedup** over sequential baselines on matrix grids up to 1024×1024.

SKILLS

Languages: C++, Python, C

AI/ML & Data: Generative AI, Natural Language Processing (NLP), RAG Pipelines, Vector Databases (ChromaDB), LLM APIs (Groq), Sentence Embeddings, Prompt Engineering

Frameworks & Libraries: Streamlit, REST APIs, NumPy, Pandas, Matplotlib

RELEVANT COURSES

Computer Science: Data Structures & Applications, Operating Systems I & II, Introduction to Programming

AI and Scientific Computing: Fundamentals of Scientific Computing, Computer-Aided Numerical Methods I & II, Computational Methods in Materials Science

Electrical: Digital Circuits, Introduction to HDL

Mathematics: Probability, Introduction to Statistics, Linear Algebra, Transform Techniques, Differential Equations, Complex Variables

POSITIONS OF RESPONSIBILITY

Collaboration & Public Relations Head, Extra Mural Lectures (EML), IIT Hyderabad

- Lead a team of 8 coordinators to secure sponsorships and promote club events.

- Manage sponsorship outreach, finances, and other official club operations.

Collaboration & Public Relations Coordinator, Extra Mural Lectures (EML), IIT Hyderabad

EXTRACURRICULAR

Volunteer, DKMS BMST Foundation India (2025)